MUHAMMAD SAQLAIN

Researcher — ML, CV, Open Source Systems, Educator

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ACHIEVEMENTS

2× Google Summer of Code (top ∼5%): Selected twice for GSoC — 2026, NumFOCUS/DeepForest (4.88%; recovering forest-damage boundaries for Oregon with University of Florida; now co-author on the DeepForest 2.0 paper, JOSS in preparation) and 2025, Google Chromium (5.43%; earned official saqlain@chromium.org status).

Open-Source Mentorship: Mentored underprivileged Pakistani students for GSoC/ESoC/LFx/OSPP via Dev Weekends; 14 selected in 2026 (9 GSoC, 1 ESoC, 2 SSoC, 2 LFx); mentees earned $50,138+ in stipends over 2 years.

Stanford Code in Place (2×): Section Leader for Stanford's Code in Place (∼16,000 students); taught CS106A to 26 students from 16 countries over 2 years. Credential

Competitive Programming: ICPC Asia Topi Regional 2024; Bronze at CALICO 2024 (UC Berkeley) (252/850+ teams); 73rd/500+ at MIT Informatics Tournament 2025; 200+ problems on LeetCode/Codeforces.

EDUCATION

University of Education — Lahore, Pakistan
MS in Information Technology — *August 2025 – 2027*
CGPA: 3.85/4.00

Coursework: Research Methodology, Advanced Information Security, Advanced Topics in Computing, Advanced Computer Networks, Mobile and Wireless Networks, Internet of Things (IoT)

Standardized Tests: IELTS — Overall 7.5 (C1 Advanced) score report

University of Education — Lahore, Pakistan
Bachelors in Information Technology — *Dec 2021 – May 2025*
CGPA: 3.13/4.00

Coursework: OOP, DBMS, Algebra, OS, DSA, System and Network Administration, AI, Computer Networks, Discrete Math, Statistics and Probability

PUBLICATIONS (IN PREPARATION)

Co-author: *DeepForest 2.0: Airborne object detection and classification for ecologists.* (Paper in preparation for the Journal of Open Source Software — Weecology Lab, University of Florida.)

RESEARCH EXPERIENCE

GSoC '26 Research Contributor — Gainesville, FL (Remote)
Weecology Lab, University of Florida – NumFOCUS / DeepForest — *May 2026 – Aug 2026*

- Mentored by **Dr. Ethan White** (PI), **Ben Weinstein** & **Josh Veitch-Michaelis** (Research Scientists), and **Henry Senyondo** (Research Software Engineer) on Recovering Historical Image Data Using Automated OrthoRegistration (4.88% selection rate).
- Tested the planned alignment approach on U.S. Forest Service Aerial Detection Survey (ADS) polygons first and showed it cannot work: no shift, rotation, or scale recovers the true boundary, because the polygons have the wrong shape, not just the wrong position. Re-scoped the project to semantic segmentation.
- Built the missing ground truth by hand (151 damage sites traced, 133 damage-free sites verified; 284 tiles of 30 cm imagery) and fine-tuned a TreeFinder-pretrained U-Net: region IoU rose from 0.115 (the survey polygon itself) to 0.268, a 2.3× gain under geography-split cross-validation.
- Found that pixel scale, not data volume, was the bottleneck: resizing every tile to 384 px made one pixel cover 0.61–3.68 m, while the model was pretrained at 0.60 m/px. Cutting fixed 0.60 m/px crops more than doubled IoU (0.116 → 0.257) and lifted zero-shot IoU from 0.040 to 0.108 with no training.
- Released the code and a final report with all 29 logged runs, including failures; running it on all ∼48,000 Oregon polygons is the next step.
- Conducted a comparative benchmarking study of DeepForest vs DeepTrees, analyzing architectural differences (RetinaNet vs U-Net+watershed), prediction outputs (bounding boxes vs polygons), vegetation indices, and fine-tuning workflows; presented findings to mentors during the selection phase.

Junior Research Assistant — Al Khobar, Saudi Arabia
Center for Artificial Intelligence (CAI), Prince Muhammad Bin Fahd University — *Jan. 2026 – Present*

- Building computer-vision pipelines for **remote sensing** and **precision agriculture**: multi-class classification, semantic segmentation, and object detection on satellite and field imagery.

- Developing deep-learning models for crop-disease diagnosis and satellite-image enhancement, benchmarking **Vision Transformer (ViT)** backbones against hybrid CNN–Transformer architectures.
- Running the data, training, and evaluation pipeline toward a first-author manuscript on remote-sensing-based crop-disease detection.

GSoC'25 Chromium OS Contributor Remote
Google Summer of Code (GSoC) – Web Audio Team *May 2025 – Nov. 2025*

- Selected for the Google Summer of Code program (5.43% selection rate) to modernize the testing infrastructure of the Chromium browser engine.
- Engineered a Generative AI automation prototype (Chromium-Automate-CLs) using Gemini models; the tool automated ∼90% of test conversion tasks and was highlighted by mentors as a roadmap for future AI-assisted workflows in Chromium.
- Designed a Python heuristic analysis system to parse 326 complex test files based on "Code Complexity Scores" (AST parsing of loops, async logic), creating the data-driven strategy for the migration.
- Delivered 96 CLs (∼95% merge rate) to migrate the Web Audio test suite to the W3C `testharness.js` framework, enforcing Google's strict JS style rules and optimizing CI bandwidth usage by ∼187 GB/day.

TEACHING EXPERIENCE (volunteer)

Section Leader (2× Selection) California, United States (Remote)
Stanford University: Code in Place *Apr 2025 – Jun 2025, Apr 2026 – Present*

- Selected twice for Stanford's Code in Place, one of the most diverse CS courses globally (∼16,000 enrolled students).
- **2026:** Selected from ∼2,000 candidates to teach CS106A Python to 11 students from 8 countries: UK (3), Venezuela (2), Serbia, Germany, Peru, Czech Republic, Russia, USA.
- **2025:** Taught Python to ∼15 international students from 11 countries: USA (5), Somalia, Thailand, Chile, Philippines, Germany, Ecuador, India, Peru, Canada. Promoted to Section Leader Mentor for 1-on-1 support and taught 6 TeachNow sessions. Credential
- Over 2 years, taught 26 students from 16 countries.

Mentor Remote
Dev Weekends *May 2024 – Present*

- Initiated and led a 6-month open-source mentoring drive, running weekly sessions, proposal reviews, and 1-on-1 guidance for 275+ students on competitive programs (GSoC, ESoC, LFx, OSPP).
- **2026:** 14 students selected (9 GSoC, 1 ESoC, 2 SSoC, 2 LFx). **2025:** 3 students selected (2 GSoC, 1 OSPP). Collectively, mentees earned over $50,000 in stipends.
- Delivered 25+ lectures on LeetCode and core DSA, teaching arrays, linked lists, trees, graphs, sorting, searching, and dynamic programming.

Trainer and Moderator Santa Clara, California Remote
iCodeGuru *Feb 2023 – Present*

- Offered volunteer teaching in Python and DSA at iCodeGuru and solved numerous LeetCode problems. All recording are here
- Taught courses on the IELTS speaking and writing modules specifically.

PROJECTS

Your JobBuddy (Final Year Project) GitHub — Sept 2024 – May 2025

- Developed a **GenAI-powered resume optimization platform** using Next.js (TypeScript) to maximize ATS (Applicant Tracking System) compatibility for job seekers.
- Integrated LLMs to implement resume parsing, automated scoring, and context-aware feedback on clarity and formatting.
- Designed a responsive interface featuring real-time document editing, multi-template support, and PDF generation.

QuillPDF Live Demo

- Engineered a **RAG-based (Retrieval-Augmented Generation)** SaaS using LangChain, Pinecone, and OpenAI to enable AI conversations with PDF documents.
- Built a high-performance full-stack architecture with Next.js, Prisma, and Tailwind CSS, supporting free/pro subscription tiers.

ClearOUT (AI Hackathon) Live — Submission

- Led a 6-member international team in the lablab.ai LangFlow Hackathon to build an AI-powered recycling/reuse assistant using image recognition and LangChain/LangFlow.

TECHNICAL SKILLS

Programming Languages: Python, JavaScript, TypeScript, C++

ML & Deep Learning: PyTorch, scikit-learn, NumPy, SciPy, Pandas, OpenCV

Computer Vision & Remote Sensing: Object Detection, Semantic Segmentation (U-Net), Image Registration, GeoPandas, Rasterio, GDAL, QGIS

Web Dev: Next.js, Node JS, Express JS

DevOps & Tools: Docker, CI/CD, GitHub Actions, AWS (RDS, S3), Git

AI: Generative AI, LLMs, RAG (LangChain, Pinecone)

Databases: MongoDB, MySQL, PostgreSQL

REFERENCES

Dr. Ethan White — Professor & PI, Weecology Lab @ University of Florida — LinkedIn

- Email: ethanwhite@ufl.edu
- Role: GSoC 2026 Mentor (NumFOCUS / DeepForest)
- Location: Gainesville, FL, United States

Dr. Ben Weinstein — Research Scientist, Weecology Lab @ University of Florida — LinkedIn

- Email: ben.weinstein@weecology.org
- Role: GSoC 2026 Mentor (NumFOCUS / DeepForest)
- Location: Gainesville, FL, United States

Dr. Josh Veitch-Michaelis — Research Scientist (Postdoc), Weecology Lab @ University of Florida — LinkedIn

- Email: j.veitchmichaelis@gmail.com
- Role: GSoC 2026 Mentor (NumFOCUS / DeepForest)
- Location: Gainesville, FL, United States

Henry Senyondo — Research Software Engineer, Weecology Lab @ University of Florida — LinkedIn

- Email: henrykironde@gmail.com
- Role: GSoC 2026 Mentor (NumFOCUS / DeepForest)
- Location: Gainesville, FL, United States

Hongchan Choi — Tech Lead & Manager, Google (Chromium) — LinkedIn

- Email: hongchan@google.com
- Web Audio / Web MIDI API Owner: Chromium Project
- Adjunct Professor: Stanford University
- Location: Mountain View, CA